ADJUSTABLE SPEED LIMITER [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

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Purpose, Outline

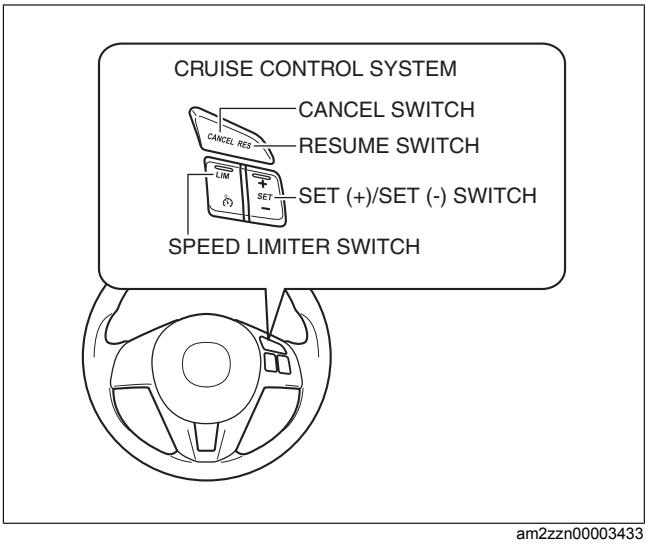
- For the purpose of safety performance improvement, the adjustable speed limiter restricts unintended excess vehicle speed by allowing the driver to optionally set the maximum vehicle speed.
- The adjustable speed limiter restricts the engine output so that the vehicle speed does not exceed the set maximum vehicle speed even if the accelerator pedal is being depressed.
- The adjustable speed limiter does not operate simultaneously with the cruise control system.

Note

- Because the brakes are not used during engine output restriction, the vehicle speed could exceed the set maximum speed limit when the vehicle is descending a slope.
- If the maximum vehicle speed exceeds the specified value, the instrument cluster flashes the maximum speed indicated in the display and activates the buzzer to warn the driver.

Function

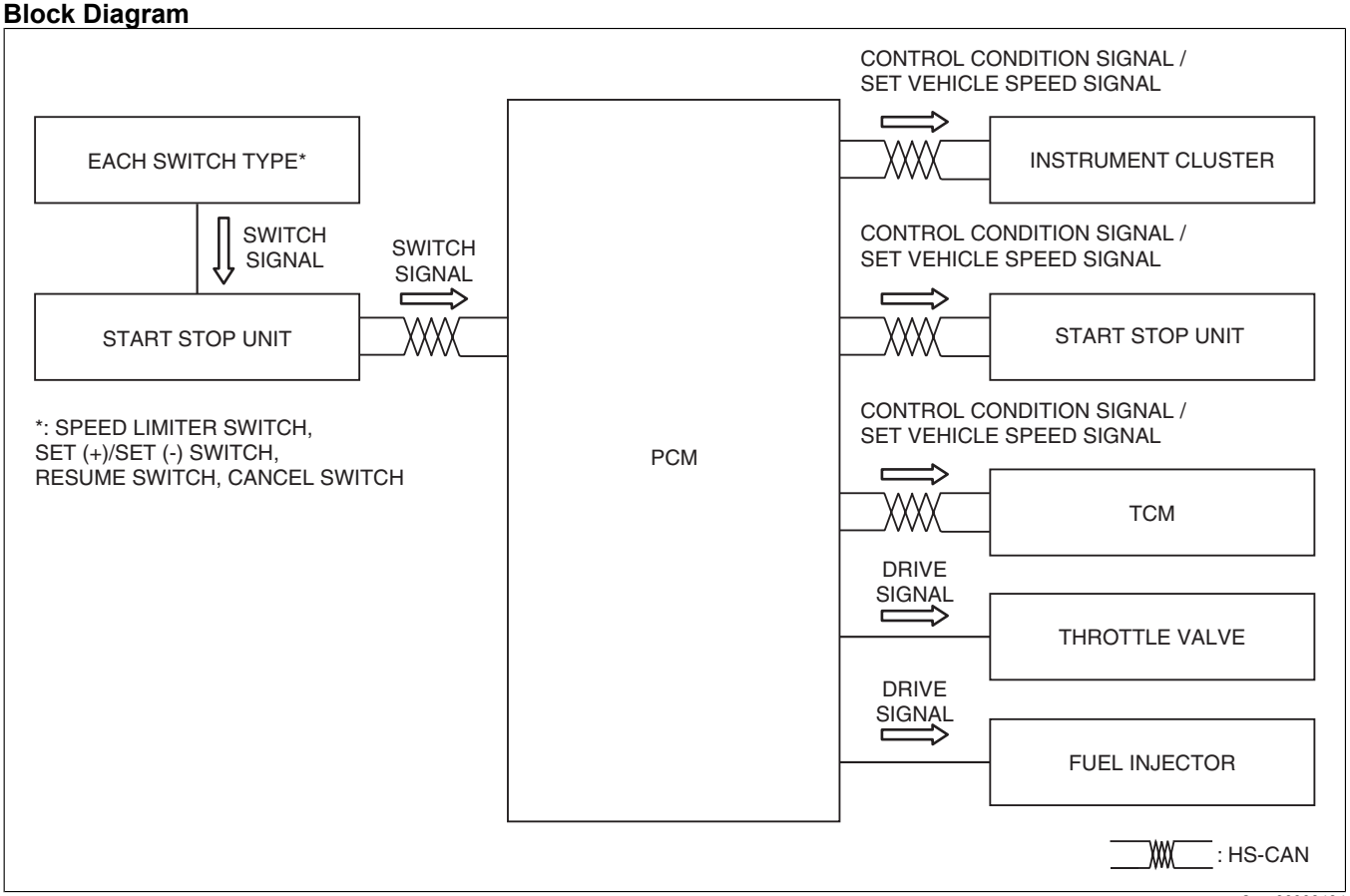
- The driver can change the adjustable speed limiter control condition by operating each type of switch on the steering wheel.



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SPEED LIMITER SWITCH	• When the speed limiter switch is pressed, the adjustable speed limiter control condition goes on maximum vehicle speed setting standby. • The standby condition can be canceled by pressing the speed limiter switch while the adjustable speed limiter is on maximum vehicle speed setting standby.
SET (+)/SET (-) SWITCH	• The maximum vehicle speed can be set. • Short-pressing the SET (+)/SET (-) switch adjusts the maximum vehicle speed in **1 km/h {0.6 mph}** increments, and long-pressing it adjusts it in **10 km/h {6.2 mph}** increments. Additionally, the maximum speed limit can be set between **30 km/h {19 mph}** and **200 km/h {124 mph}**.
RESUME SWITCH	• The maximum vehicle speed can be set to the previous maximum vehicle speed recorded by the PCM.
CANCEL SWITCH	• If the CANCEL button is pressed while the adjustable speed limiter control is operating, the adjustable speed limiter control is temporarily canceled.

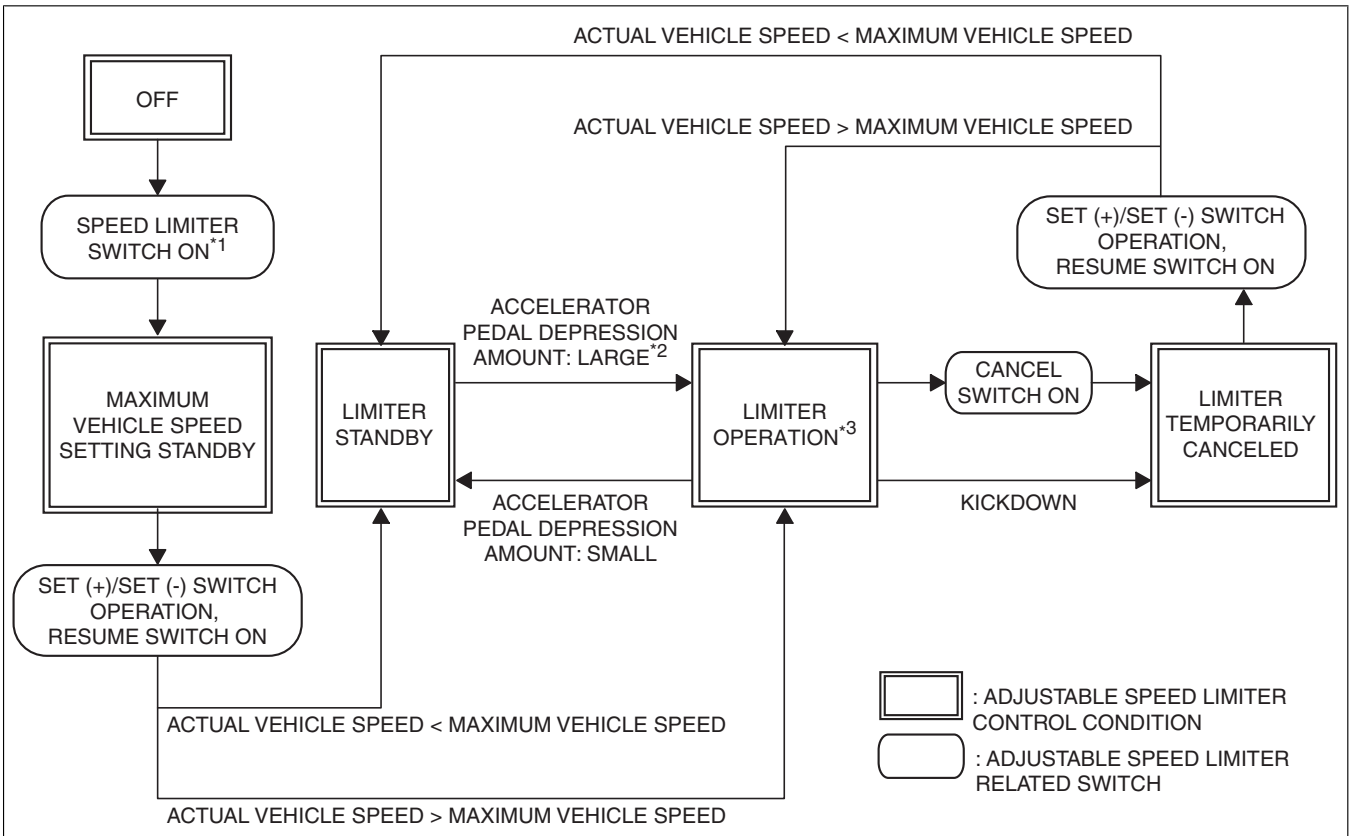
Block Diagram



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Operation

- The adjustable speed limiter control condition switches according to the driving conditions/each type of switch operation.



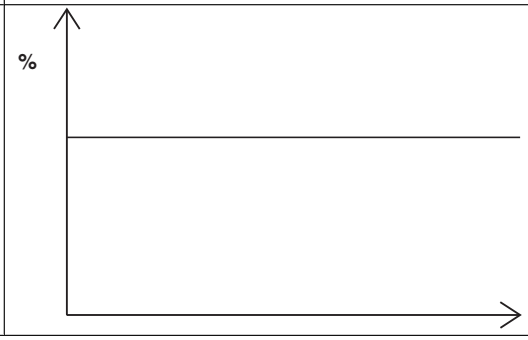
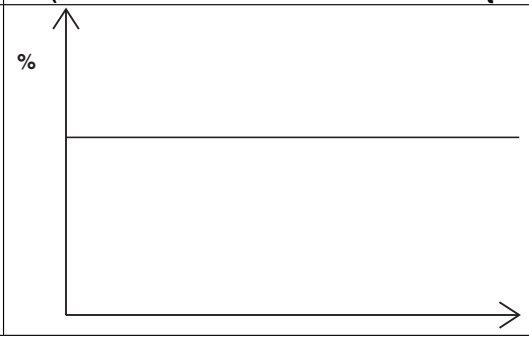
*1: IF THE ADJUSTABLE SPEED LIMITER CONTROL CONDITION IS OTHER THAN OFF, THE CONTROL CONDITION BECOMES OFF BY PRESSING THE SPEED LIMITER SWITCH.

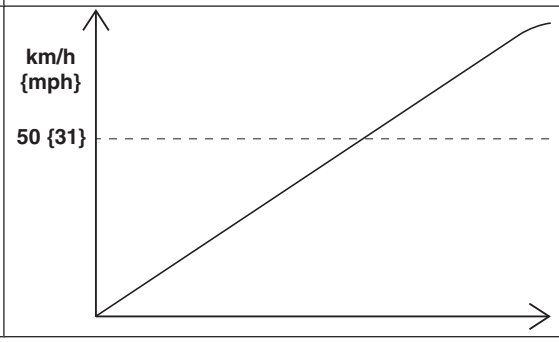
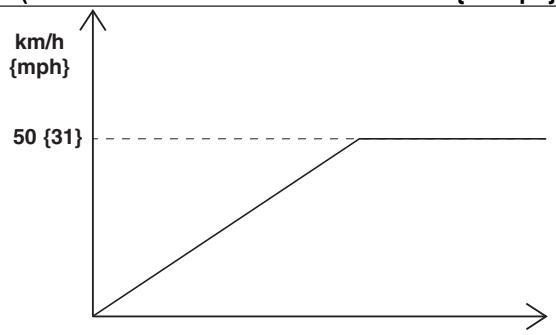
*2: DURING KICKDOWN, THE ADJUSTABLE SPEED LIMITER CONTROL CONDITION BECOMES "LIMITER TEMPORARY CANCEL".

*3: IF THE ACTUAL VEHICLE SPEED EXCEEDS THE MAXIMUM VEHICLE SPEED BY 3 km/h OR MORE, THE INDICATOR FLASHES AND THE BUZZER IS ACTIVATED TO WARN THE DRIVER.

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- When the variable speed limiter control transitions to a limiter temporarily canceled condition due to kickdown, if the actual vehicle speed exceeds the maximum vehicle speed by 3 km/h or more, the maximum vehicle speed flashes.
- When the CANCEL switch is pressed, the variable speed limiter control is temporarily stopped.
- While the adjustable speed limiter control is operating, the PCM restricts the engine output so that the maximum vehicle speed is not exceeded even if the accelerator pedal is depressed.

	NORMAL DRIVING	**DURING ENGINE OUTPUT RESTRICTION (MAXIMUM VEHICLE SPEED: 50 km/h {31 mph}**
ACCELERATOR PEDAL OPENING ANGLE SIGNAL		

	NORMAL DRIVING	DURING ENGINE OUTPUT RESTRICTION (MAXIMUM VEHICLE SPEED: 50 km/h {31 mph}
VEHICLE SPEED		

- If the actual vehicle speed while descending a slope exceeds the maximum vehicle speed by **3 km/h {2 mph}** or more, the PCM sends a adjustable speed limiter warning alarm activation request to the instrument cluster.The PCM cancels the activation request to the instrument cluster if the actual vehicle speed is decreased **3 km/h {2 mph}** lower than the maximum vehicle speed.
- The instrument cluster receives the activation request from the PCM and activates the adjustable speed limiter warning alarm.

Engine output restriction method

- The PCM restricts the engine output using the following controls.
  - — Drive-by-wire control
    - Engine output is restricted by adjusting the throttle valve opening angle.
  - — Fuel injection control
    - Engine output is restricted by performing fuel-cutting.
- The TCM shifts the gears optimally according to the control condition signal/set vehicle speed signal from the PCM.

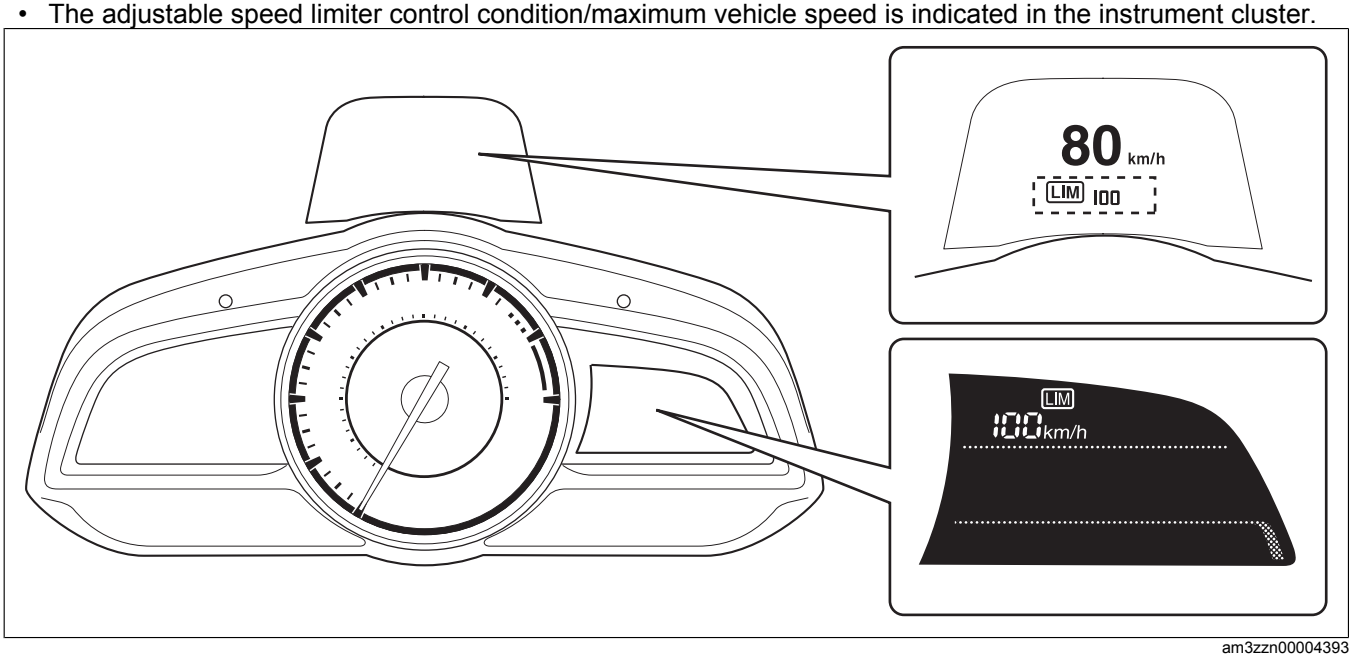
Engine output restriction cancelation

- While the adjustable speed limiter control is operating, the engine output restriction can be canceled by performing the following operations.
  - — CANCEL switch is ON
  - — Kickdown
  - — Speed limiter switch is OFF*
  - — Engine switch is OFF*

* : The adjustable speed limiter control condition is OFF.

Instrument cluster display

- The adjustable speed limiter control condition/maximum vehicle speed is indicated in the instrument cluster.



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- The content displayed in the instrument cluster based on the adjustable speed limiter control condition is as follows.

ADJUSTABLE SPEED LIMITER CONTROL CONDITION	INSTRUMENT CLUSTER DISPLAY	NOTE
OFF		—
MAXIMUM VEHICLE SPEED SETTING STANDBY		—
LIMITER STANDBY		—
LIMITER OPERATION (ACTUAL VEHICLE SPEED < MAXIMUM VEHICLE SPEED)		—
LIMITER OPERATION (ACTUAL VEHICLE SPEED > MAXIMUM VEHICLE SPEED)		• The maximum vehicle speed “100" flashes, and the adjustable speed limiter warning alarm activates.
LIMITER TEMPORARILY CANCELED		• When the variable speed limiter control transitions to a limiter temporarily canceled condition due to kickdown, if the actual vehicle speed exceeds the maximum vehicle speed by 3 km/h or more, the maximum vehicle speed “100” flashes.